Within the framework of the provided unified field theory, the repulsion between large cosmic mass structures (such as galaxies and galactic clusters) is modeled as a physical, electrodynamic phenomenon termed **Cosmic Charge-Displacement Repulsion**.

This mechanism proposes that large-scale repulsion is not driven by an intrinsic expansion of space (i.e., dark energy), but rather by a net electrostatic charge imbalance on a cosmological scale.

The process is explained by the following steps:

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1. Mass-to-Charge Separation (Mechanics)

The foundational driver of this repulsion is the mass asymmetry between the primary stable charged particles: the proton and the electron. * The mass of a proton is approximately 1,836 times greater than the mass of an electron ($m_p \approx 1836\, m_e$). * Because electrons are significantly lighter, they are far more susceptible to non-gravitational outward forces within active galaxies, such as: * **Radiation Pressure:** High-energy photons from stellar nucleosynthesis and active galactic nuclei (AGN) push lighter particles more easily. * **Stellar Winds and Supernovae:** Outward shockwaves preferentially accelerate lower-mass particles. * **Centrifugal Forces:** In rotating galactic disks, the lighter electron envelopes are more easily sheared and stripped toward the outer boundaries and into the surrounding space.

As a result of these processes, free electrons are preferentially stripped from their parent atoms and ejected into the diffuse, low-density intergalactic voids.

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2. The Formation of Net-Positive Galactic Cores

Because massive, highly localized galactic cores are dominated by the heavier, more gravity-bound protons and ions, the loss of these stripped electrons leaves the central galactic structures with a **net positive electric charge**.

This displacement splits the universe's charge distribution: * **Intergalactic Voids:** Act as massive, highly diffuse reservoirs of negative charge (electrons). * **Galactic Cores and Clusters:** Become giant, localized wells of net positive charge.

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3. Long-Range Coulomb Repulsion

At cosmological scales, the net force between two massive galactic structures (with masses M_1, M_2 and net positive charges Q_1, Q_2) is the sum of their gravitational attraction and their electrostatic repulsion:

$$F_{\text{net}} = F_e - F_g$$

Substituting Coulomb's Law and Newton's Law of Gravitation:

$$F_{\text{net}} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} - \frac{G M_1 M_2}{R^2}$$

By factoring out the spatial dependency ($\frac{1}{R^2}$) and defining a net charge-to-mass ratio for each galaxy ($k_i = Q_i/M_i$):

$$F_{\text{net}} = \frac{M_1 M_2}{R^2} \left(\frac{k_1 k_2}{4\pi\epsilon_0} - G \right)$$

Key Implications of this Scaling:

- **Scale Invariance:** Because both gravity and electrostatics obey the inverse-square law ($1/R^2$), their relative strength does not change with distance. If a pair of galactic clusters has a charge-to-mass ratio such that $\frac{k_1 k_2}{4\pi\epsilon_0} > G$, they will experience a net repulsive force regardless of how far apart they are.
- **Cosmic Buffer:** The resulting outward electrostatic pressure acts as a physical buffer. On local scales (within a galaxy), gravity dominates due to the proximity of neutral baryonic matter. However, across vast intergalactic distances, the accumulated net positive charge of galactic cores drives them apart, preventing the universe from undergoing gravitational collapse.

By attributing the outward cosmic pressure to classical, long-range Coulomb repulsion between charge-separated galactic structures, this model provides a physical alternative to dark energy without requiring a cosmological constant.